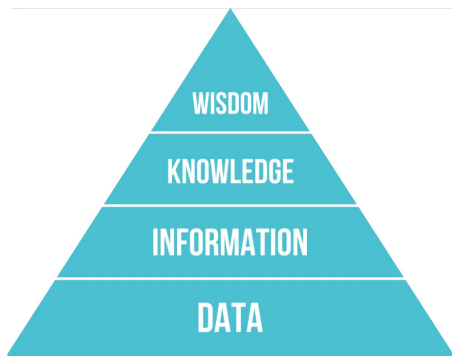
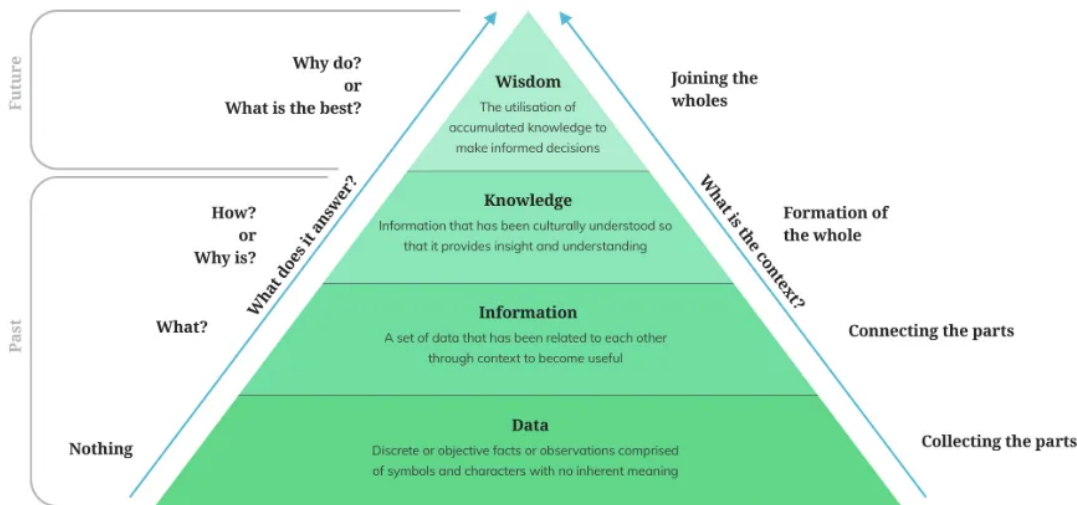


Ch0

1. Data, information, knowledge, wisdom



**Data, information, knowledge and wisdom
(DIKW) pyramid**



- Data are **raw, unstructured** facts and figures.
- Information provides **context**, meaning, and purpose to data by organizing it into a coherent 有条理的 form.
- Knowledge is accumulated information that **reveals patterns** and connections, offering insights and understanding.
- Wisdom is the **application** of knowledge to make sound judgments, foresee outcomes, and guide actions in a thoughtful and beneficial way.

2. Type of data

Structured Data

- Sensor data
- Weblog data
- Financial data
- Click-stream data
- Point of sale data
- Accounting data

Unstructured Data

- Satellite images
- Photographic data
- Video data
- Social media data
- Text message
- Voice mail data

Unstructured Data – image/video

Structured Data

- **Click-stream**: following
 - Point of sale: cashier
-

3. Model related to Big data

Computing model – 决定了计算资源如何被使用、数据如何被处理，以及程序如何运行。

Big data refers to enormous, complex, and rapidly growing datasets that traditional data processing software cannot handle

对“大数据”发展非常关键的计算模型：

- **Distributed computing** – Processes and manages algorithms across many machines in a computing environment
 - **分布式计算**：指的是将一个计算任务拆分后，分配到多个计算机上同时运行。
 - 让多个机器协同处理数据，提升处理速度和系统稳定性
 - **Virtualization** – The creation of a virtual (rather than actual) version of computing resources, such as an operating system, a server, a storage device, or network resources
 - 是指用软件技术把真实的硬件资源“虚拟”成多个独立的资源单元
 - 虚拟机
 - 提高资源利用率，灵活部署和管理计算环境
-

4. Business Focus Areas of Big Data



4.1. Data mining

- **Definition:** The process of analyzing data to **extract information** not offered by the raw data alone
 - Data mining is the process of extracting useful and previously **unknown patterns** from large datasets
 - Transform raw data into meaningful and understandable information
 - 通过算法、统计方法或机器学习，**挖掘出隐藏的模式、趋势或洞察。**

- **3 main elements of data mining:**

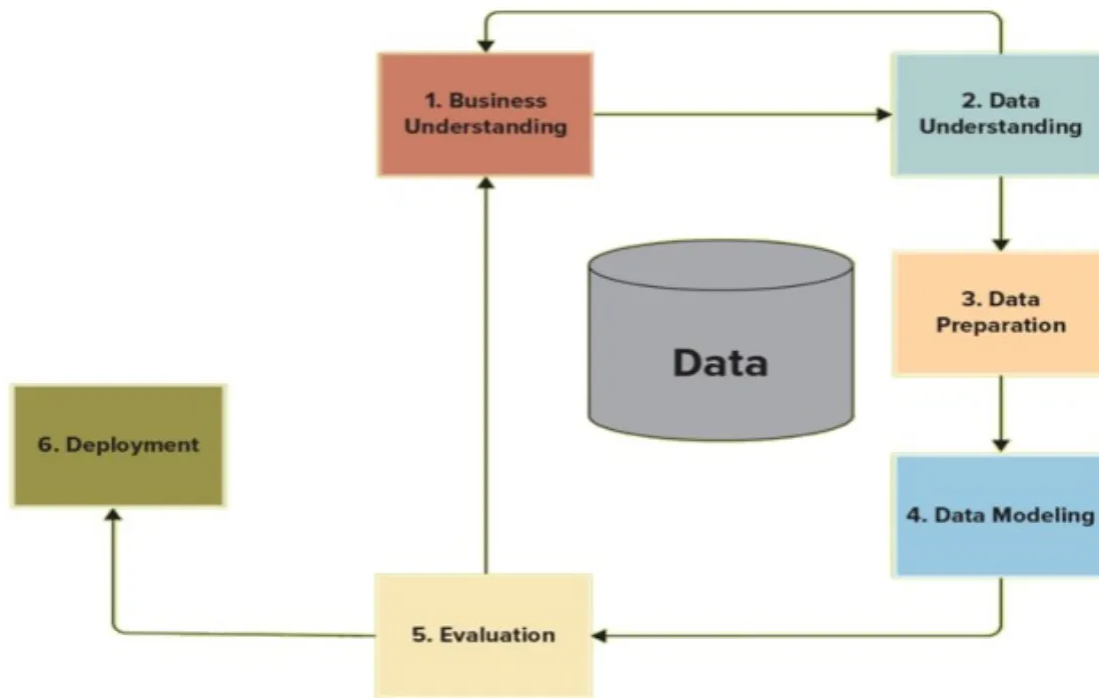
- Data
- Discovery – identifying **new** patterns, trends ,and insights
- Deployment – implementing discoveries 把发现的结果应用到实际业务

Data: 收集用户点击、浏览、购买等行为数据；

Discovery: 用算法发现用户在晚上更倾向于购买某类商品；

Deployment: 根据这个发现调整推荐系统，在晚上推送相关商品，提高转化率。

- **Data Mining process Model**



1. **Business Understanding** – identifying the business problem, setting goals, and KPIs.
 - Define the business objectives, constraints, and success criteria.
2. **Data understanding** – 探索和分析已有数据，判断数据质量、结构和潜在问题 – This step identifies data sources, detects quality issues, and uncovers patterns or relationships. It ensures the data is suitable for analysis.
3. **Data preparation** – Data with low quality / Different format data.
 - Raw data → Collect, clean, and label raw data in a format suitable → for machine learning (ML) algorithms, followed by data exploration and visualization
4. **Data Modeling** – 应用统计或机器学习模型来分析数据 Apply algorithms (e.g., classification, regression, clustering) to extract patterns.
5. **Data Evaluation**
6. **Deployment** – 将模型或分析结果应用到实际业务中 Integrate the model into business processes

第 1 步（业务理解）是决定你要盖什么房子（别墅？公寓？仓库？）

第 2 步（数据理解）就是去看你手上有哪些建材，质量如何，能不能用来盖你想要的房子

• Data Mining ANALYSIS TECHNIQUES

Data Mining 分析技术：

Data profiling - The process of **collecting** statistics and information about data in an existing source

Data replication - The process of **sharing information** to ensure consistency between multiple data sources

Recommendation engine - A data mining algorithm that analyzes a customer's purchases and actions on a website and then uses the data to **recommend** complementary products

1. Data Profilling （数据剖析）

对现有数据源中的**统计数据和信息**进行**收集**的过程

分析已有数据源的结构、质量和分布。

how many? average?

清洗数据前了解字段缺失率、异常值、重复项等

2. Data replication （数据复制）

在**多个数据源**之间同步数据，确保**一致性**。

3. Recommendation engine （推荐引擎）

分析用户行为，推荐相关产品或内容。

这些技术不是直接用于挖掘数据模式，而是用于**支撑数据挖掘的环境和应用场景**：

技术	作用	分类定位
Data profiling	检查数据质量、统计特征	数据准备阶段
Data replication	保证多个数据源一致性	数据工程/系统维护
Recommendation engine	用挖掘结果驱动应用（如推荐）	应用层算法（非通用分析方法）

📖 这些技术更偏向“**数据挖掘的外围支持**”，或者“**挖掘结果的应用**”

经典的数据挖掘算法（建模与模式识别）**建模技术**：

1. Classification （分类）

Assigns records to one of a **predefined set** of classes

2. Estimation （估计）

Determines values for an unknown continuous variable behavior or **estimated future value** – 预测连续变量的值

3. **Affinity Grouping**（关联分组）

Determines which things go together

- 购物篮分析

4. **Clustering**（聚类）

将数据划分为若干相似子群体。

Segments a heterogeneous population of records into a number of more homogeneous subgroups

维度	分类 (Classification)	聚类 (Clustering)
是否有标签	✅ 有 (监督学习)	❌ 无 (非监督学习)
输入数据	已知类别的数据样本	未知类别的数据样本
输出结果	为每个样本分配一个已有的类别	将样本划分为若干组 (类别未知)
目标	学习如何预测标签	发现数据中的结构或相似性
应用场景	垃圾邮件识别、疾病诊断、图像分类	客户细分、市场分析、异常检测

Data mining tools – Use a variety of techniques to find **patterns** and **relationships** in large volumes of information that predict future behavior and guide decision making

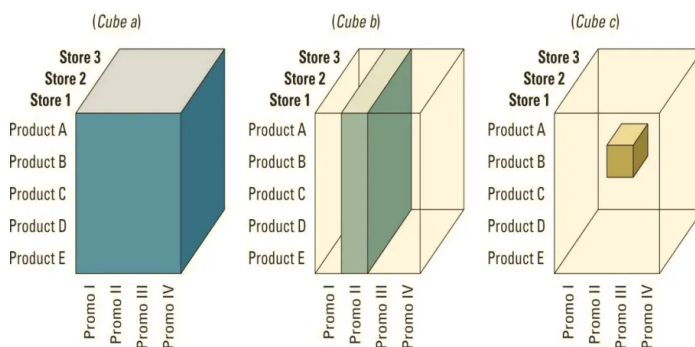
- 应用场景
 - Building budgets and other financial information
 - Finding the best customers who spend the most money – 找出最有价值的客户
 - Detecting fraud by identifying **deceptive spending patterns** – 识别欺诈性消费模式
 - **欺诈性消费模式**是指那些与正常用户行为显著不同、可能暗示欺诈行为的消费行为或交易特征。
 - Unusual amounts: sudden large purchases
 - Frequent transactions in a short time
 - Geographic anomalies: rapid spending across different regions
 - High-risk categories: buying resellable luxury or electronics
 - Financial loss; Regulatory risk; Increased costs:
 - Keeping customers from leaving or migrating to competitors – 防止客户流失或转向竞争对手
 - Promoting and hiring employees to ensure success – 提升和招聘员工以确保成功

Prediction Modeling Techniques 预测建模技术

- Optimization modeling – Finding the best solution among many options
- Forecasting modeling – Predicting future trends using time-series data
- Regression modeling – 预测连续变量 continuous variables的数值，如房价或用户活跃度

4.2. Data analysis

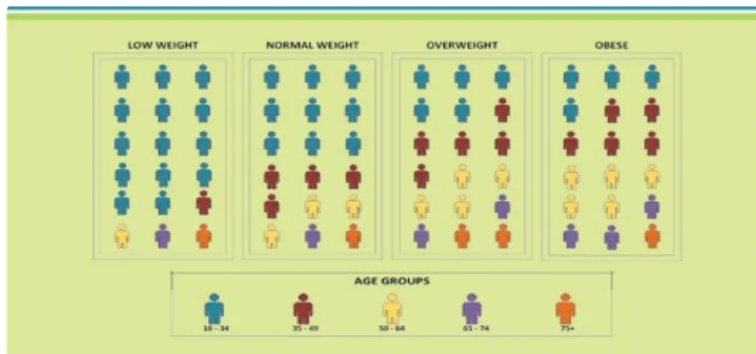
- Information cube -> representation of multidimensional information



- Algorithms – Mathematical formulas placed in software that performs an analysis on a data set
 - Uncover:
 - Outliers 异常值
 - Analysis paralysis 分析瘫痪
 - 分析瘫痪是指个体陷入过度分析或过度思考的情绪状态，导致迟迟无法做出决策或采取行动，从而使结果停滞不前
 - **Analysis paralysis** – User goes into an emotional state of over-analysis (or over-thinking) a situation so that a decision or action is never taken, in effect paralyzing the outcome
- Fast data – The application of big data analytics to smaller data sets in **near-real or real-time** to solve problems or create business value.
 - 指将大数据分析技术应用于小型数据集，并在近实时或实时环境中处理，以快速响应业务需求
- 八种关键分析方法:
 - Behavioral Analysis 行为分析
 - Correlation Analysis 相关性分析 – 识别变量之间的关系

- ### 4.3. Data visualization

- a visual representation of data or complex information, using graphics, charts, and text to make complex ideas easier to understand, spot trends, and communicate quickly and clearly



Data visualization tools - Move beyond **Excel graphs** and **charts** into sophisticated analysis techniques such as pie charts, controls, instruments, maps, time-series graphs, and more



1. What are the six steps in the data-mining process and why is each important?

Cross-Industry Standard Process for Data Mining – CRISP-DM

Business Understanding

Data Understanding

Data preparation – better model performance and interpretability.

Data modeling

Evaluation

Deployment – Without deployment, even the best model is just academic.

2. What are data-mining techniques? Provide examples of how you would use each one in business.

Affinity grouping/ Classification / Estimation/ Clustering

3. What is data-driven decision management?

- Management is the act of coordinating resources, planning, organizing, leading, and controlling tasks to achieve an organization's goals.

By using the data, which is the fact to help coordinating resources, planning, organizing, leading, and controlling tasks to achieve an organization's goals.

User – Analysis paralysis BUT! DATA WON'T Data is objective

Guide the decision-making process with data rather than intuition or experience.

4. What are the four data-mining techniques for predictions and why are they important to a business?

- Optimization modeling
- Forecasting modeling
- Regression modeling